

Werner Bisschoff

EMBEDDED SYSTEMS & REAL-TIME SOFTWARE ENGINEER

Cape Town, Western Cape, South Africa

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Summary

Rigorous Systems Engineer with 5+ years of experience engineering real-time, event-driven software architectures (C/C++, FreeRTOS) and secure, deterministic communication protocols. Specializes in designing robust firmware, high-performance IPC mechanisms, and hardware-in-the-loop simulation frameworks. Governed by strict Spec-Driven Development (SDD) methodologies and automated validation pipelines to eliminate runtime faults in production environments.

Experience

Divergent Tabletop

Cape Town

PRINCIPAL SYSTEMS ARCHITECT

Jul 2025 – Present

- Engineered and deployed a distributed knowledge-management infrastructure
- Integrated a CRDT synchronization layer (**y_ex**) to handle real-time modifications; implemented an actor-based buffering model that reduced state-sync latency to sub-50ms thresholds.
- Designed a dynamic BEAM supervision tree with a **:one_for_one** strategy to isolate concurrent page editing instances, ensuring 99.9% fault isolation and zero system-wide cascading crashes.
- Implemented a threshold-based buffer flush and scheduled heap hibernation (**hibernate_after**), reducing process memory footprints by ~60% under active network I/O concurrency.
- Applied Monotropic focus principles and Universal Design to minimize user information processing overhead and ensure highly predictable, deterministic information layout.

FARO Africa

Cape Town

EMBEDDED SYSTEMS & INTEGRATION ENGINEER

Aug 2024 – Nov 2025

- Designed ISO 14443-4 NFC communication architectures for e-paper display nodes and secure card reader matrices.
- Engineered lock-free asynchronous device management services bridging hardware peripherals with application layers.
- Built Python-based hardware peripheral simulators to catch edge-case device state corruptions prior to flashing.
- Introduced LLM-assisted development workflows to improve debugging speed and code review throughput.

Ingenics Digital GmbH (through ViVa Outsourcing)

Remote Work

FIRMWARE ENGINEER (CONTRACT)

Mar 2023 – May 2024

- Architected deterministic finite state machines (FSM) on dual-core ESP32 using FreeRTOS for concurrent sensor arrays.
- Developed Cap'n Proto inter-process communication over serial lines, achieving sub-millisecond serialization latency.
- Implemented strict compile-time and runtime hardware state guards via rigorous assertions preventing fragmentation.

UMAN Technologies

Century City, Cape Town

SOFTWARE DEVELOPER

Mar 2021 – Dec 2022

- Implemented RPC services using SOME/IP protocol and reduced performance bottlenecks with **perf**.
- Built IPC/RPC layers in C++ and Python using Cap'n Proto; designed node trees exposing process variables via IPC interface.

North-West University

Potchefstroom

JUNIOR LECTURER

Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students.

Education

North-West University

Potchefstroom

B.ENG. COMPUTER AND ELECTRONIC ENGINEERING

2020

- Focus on embedded systems, software engineering, and electronic design
- Developed NFC payment protocol emulator (Kotlin/Android), STM32 firmware (C), and PID motor controller (Arduino)

Technical Application Scope

Primary	C++ FreeRTOS ESP32/ESP-IDF NFC APDU (ISO 14443-4)
Secondary	NimBLE UART/SPI/I2C TinyFrame Hardware-in-the-Loop (HIL) Testing
Foundational	Assembly Digital Logic System Modeling
Other	ERPNext (Integration-focused) Expo Mobile